WORKED SOLUTION

STEP III 2021, Question 6

Pure • Inverse Functions and Derivatives

STEP III 2021, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Both functions look forbidding, but their derivatives turn out to be very simple. A function whose derivative is $\frac{1}{1+x^2}$ must be $\tan^{-1} x$ plus a constant, and a function with zero derivative is constant, but only on each interval where it is defined. So the real work is to find the constant on each separate piece, using a convenient value or a limit, and to watch the jumps at the points where the function is undefined.

Solution

Part (i)

Write $t = \tan\alpha$, so $t > 0$, and let $u = \dfrac{xt+1}{t-x}$, so $f_\alpha(x) = \tan^{-1} u$. By the quotient rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{t(t-x)+(xt+1)}{(t-x)^2} = \frac{1+t^2}{(t-x)^2},$$

and

$$1+u^2 = \frac{(t-x)^2+(xt+1)^2}{(t-x)^2} = \frac{t^2+x^2+x^2t^2+1}{(t-x)^2} = \frac{(1+t^2)(1+x^2)}{(t-x)^2}.$$

So by the chain rule

$$f'_\alpha(x) = \frac{1}{1+u^2}\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2},$$

as required.

? **Ask yourself** Does this mean $f_\alpha(x) = \tan^{-1} x + c$ for a single constant c?

No. The function is undefined at $x = \tan\alpha$, so the domain is two separate intervals, and on each one $f_\alpha(x) - \tan^{-1} x$ has zero derivative and so is constant, but the two constants need not agree.

For $x < \tan\alpha$: at $x = 0$ we have $f_\alpha(0) = \tan^{-1}(\cot\alpha) = \frac{1}{2}\pi - \alpha$, since $0 < \frac{1}{2}\pi - \alpha < \frac{1}{2}\pi$. Hence

$$f_\alpha(x) = \tan^{-1} x + \tfrac{1}{2}\pi - \alpha \qquad (x < \tan\alpha).$$

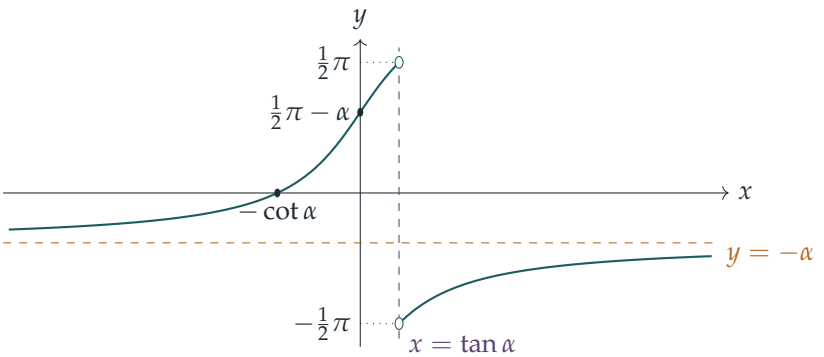
For $x > \tan\alpha$: as $x \to \tan\alpha$ from above, the numerator $xt + 1$ tends to $1 + t^2 > 0$ while the denominator $t - x$ tends to 0 through negative values, so $u \to -\infty$ and $f_\alpha(x) \to -\frac{1}{2}\pi$. Since $\tan^{-1} x \to \alpha$, the constant is $-\frac{1}{2}\pi - \alpha$:

$$f_\alpha(x) = \tan^{-1} x - \tfrac{1}{2}\pi - \alpha \qquad (x > \tan\alpha).$$

(The same limit argument from below gives $u \to +\infty$ and the value $\frac{1}{2}\pi$, agreeing with the first formula.)

So the graph is the graph of $y = \tan^{-1} x$ in two pieces, one shifted up by $\frac{1}{2}\pi - \alpha$ and the other shifted down by $\frac{1}{2}\pi + \alpha$. Its features are:

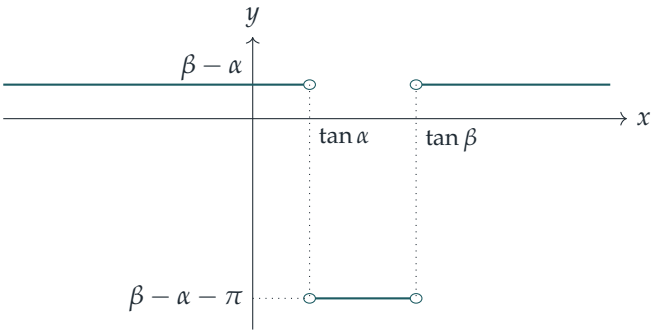
- it is increasing on each piece, with gradient $\frac{1}{1+x^2}$;
- as $x \to \pm\infty$, $y \to -\alpha$ (from above on the left, from below on the right), so $y = -\alpha$ is an asymptote at both ends;
- the left piece runs from $-\alpha$ up to $\frac{1}{2}\pi$ (not attained) at $x = \tan\alpha$, crossing the y-axis at $\frac{1}{2}\pi - \alpha$ and the x-axis where $\tan^{-1} x = \alpha - \frac{1}{2}\pi$, that is at $x = -\cot\alpha$;
- the right piece starts just above $-\frac{1}{2}\pi$ at $x = \tan\alpha$ and rises to $-\alpha$; it lies entirely below the x-axis;
- at $x = \tan\alpha$ there is a jump of $-\pi$, and both pieces have gradient $\cos^2\alpha$ there.



The difference. Since $0 < \alpha < \beta < \frac{1}{2}\pi$ we have $\tan\alpha < \tan\beta$, so there are three intervals. Using the two formulas for each function:

- for $x < \tan\alpha$: $f_\alpha(x) - f_\beta(x) = (\frac{1}{2}\pi - \alpha) - (\frac{1}{2}\pi - \beta) = \beta - \alpha$;
- for $\tan\alpha < x < \tan\beta$: $f_\alpha(x) - f_\beta(x) = (-\frac{1}{2}\pi - \alpha) - (\frac{1}{2}\pi - \beta) = \beta - \alpha - \pi$;
- for $x > \tan\beta$: $f_\alpha(x) - f_\beta(x) = (-\frac{1}{2}\pi - \alpha) - (-\frac{1}{2}\pi - \beta) = \beta - \alpha$.

The $\tan^{-1} x$ terms cancel, so the graph is a step function: the horizontal line $y = \beta - \alpha$ (which lies between 0 and $\frac{1}{2}\pi$), except on the interval between $\tan\alpha$ and $\tan\beta$, where it drops by π to $y = \beta - \alpha - \pi$ (which lies between $-\pi$ and $-\frac{1}{2}\pi$). The function is undefined at $x = \tan\alpha$ and $x = \tan\beta$.



Part (ii)

Differentiate each term. Since $\dfrac{\mathrm{d}}{\mathrm{d}s}\tanh^{-1} s = \dfrac{1}{1-s^2}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\tanh^{-1}(\sin x) = \frac{\cos x}{1-\sin^2 x} = \frac{1}{\cos x} = \sec x,$$

and since $\dfrac{\mathrm{d}}{\mathrm{d}s}\sinh^{-1} s = \dfrac{1}{\sqrt{1+s^2}}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\sinh^{-1}(\tan x) = \frac{\sec^2 x}{\sqrt{1+\tan^2 x}} = \frac{\sec^2 x}{\sqrt{\sec^2 x}} = \frac{\sec^2 x}{|\sec x|} = |\sec x|.$$

The square root is the positive one, which is why the modulus appears. So $g'(x) = \sec x - |\sec x|$. For $\frac{1}{2}\pi < x < \frac{3}{2}\pi$ we have $\cos x < 0$, so $|\sec x| = -\sec x$ and

$$g'(x) = 2\sec x,$$

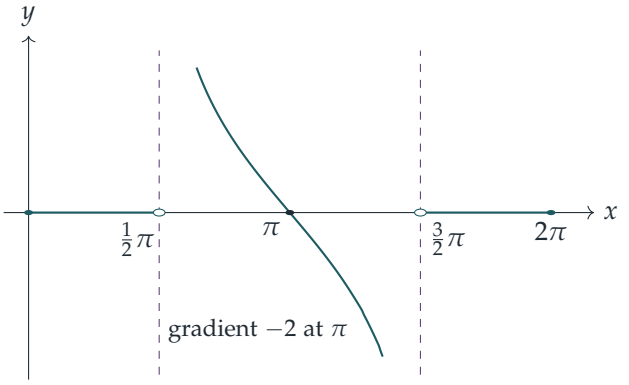
as required. On the other two intervals, $0 \le x < \frac{1}{2}\pi$ and $\frac{3}{2}\pi < x \le 2\pi$, we have $\cos x > 0$, so the same calculation gives $g'(x) = 0$.

The outer intervals. Here g is constant on each interval. Since $g(0) = \tanh^{-1} 0 - \sinh^{-1} 0 = 0$ and likewise $g(2\pi) = 0$, we get $g(x) = 0$ for $0 \le x < \frac{1}{2}\pi$ and for $\frac{3}{2}\pi < x \le 2\pi$.

The middle interval. Here $g'(x) = 2\sec x < 0$, so g is strictly decreasing. At the midpoint, $g(\pi) = \tanh^{-1} 0 - \sinh^{-1} 0 = 0$, and $g'(\pi) = 2\sec\pi = -2$. At the ends:

- as $x \to \frac{1}{2}\pi^+$, $\sin x \to 1$ so $\tanh^{-1}(\sin x) \to +\infty$, while $\tan x \to -\infty$ so $-\sinh^{-1}(\tan x) \to +\infty$; hence $g(x) \to +\infty$;
- as $x \to \frac{3}{2}\pi^-$, $\sin x \to -1$ and $\tan x \to +\infty$, so both terms send $g(x) \to -\infty$.

Also $g''(x) = 2\sec x \tan x$. For $\frac{1}{2}\pi < x < \pi$ both $\sec x$ and $\tan x$ are negative, so $g'' > 0$; for $\pi < x < \frac{3}{2}\pi$, $\sec x < 0 < \tan x$, so $g'' < 0$. So $(\pi, 0)$ is a point of inflection with gradient -2, and the curve has rotational symmetry about it (indeed $g'(2\pi - x) = g'(x)$ and $g(\pi) = 0$). In fact, since g and $2\tanh^{-1}(\sin x)$ have the same derivative on this interval and both vanish at π, $g(x) = 2\tanh^{-1}(\sin x) = 2\ln|\sec x + \tan x|$ there.



Checks

- With $\alpha = 0.6$: $f_\alpha(0) = \tan^{-1}(\cot 0.6) \approx 0.9708 = \frac{1}{2}\pi - 0.6$, and $f_\alpha(\pm 10^9) \approx -0.6$, as the asymptote predicts.
- $g(1) = \tanh^{-1}(\sin 1) - \sinh^{-1}(\tan 1) = 1.2261 - 1.2261 = 0$, confirming the flat pieces. A computer check of thousands of random points matched all the formulas above to within 10^{-6}.

Common slips

- Writing $f_\alpha(x) = \tan^{-1} x + c$ with one constant everywhere. Zero derivative only gives a constant on an interval.
- Writing $\sqrt{\sec^2 x} = \sec x$. It is $|\sec x|$, and that sign is the whole reason g is zero on two intervals and not on the third.

Beyond STEP

That zero derivative on an interval forces a constant is a consequence of the Mean Value Theorem, and it fails on sets that are not intervals, as both parts show. Part (i) is the addition formula for $\tan^{-1}$ with its correction by $\pm\pi$, the kind of branch bookkeeping that is central in complex analysis. Abbott's *Understanding Analysis* gives a careful account of the Mean Value Theorem and its consequences.

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